

IE 306 — Systems Simulation

Assignment 2: Coordinated Traffic Corridor Simulation

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1 Model Logic and Assumptions

We modeled the corridor as a discrete-event system in **SimPy** with two intersections, A (upstream) and B (downstream), connected by a finite-capacity point-queue link (capacity $N_{\max} = 40$ vehicles). Both intersections follow fixed-time two-phase control:

N/S green (45 s) $\rightarrow$ all-red (4 s) $\rightarrow$ W/E green (37 s) $\rightarrow$ all-red (4 s).

The three scenarios were implemented exactly as assigned:

- S1:** Uncoordinated, no emergencies ($\phi = 0$ s)
- S2:** Coordinated, no emergencies ($\phi = 20$ s)
- S3:** Coordinated, with emergencies ($\phi = 20$ s)

Arrival processes. South-entry flow at A is Poisson with rate $\lambda = 900$ veh/h and class split (car/bus/emergency) according to assigned proportions; W $\rightarrow$ E at A and B are independent Poisson processes with $\lambda_{WE} = 400$ veh/h each.

Turning logic. N/S vehicles turn east at A with probability 0.30 and at B with probability 0.20.

Bus priority. Buses are non-preemptive priority in approach queues (ahead of regular cars, but never interrupt in-service vehicles).

Emergency preemption. If emergency arrives during non-N/S phase, controller immediately inserts all-red 4 s, serves N/S for at least 10 s, then resumes regular cycle from interruption point remainder.

Downstream blocking. At A, non-emergency N/S vehicles can enter link only if signal is N/S green and link occupancy $< N_{\max}$. Emergencies bypass this occupancy gate, as specified.

Service and delay definition. Service time at each approach is exponential with mean 2 s. Delay is queue waiting time before service starts. Through-delay is cumulative approach waiting for vehicles that enter south and exit north at B.

Travel time. Link travel time follows

$$t_{travel}(n) = 20 \cdot \frac{1}{1 - n/40},$$

with a small multiplicative noise stream (mean 1.0, SD 0.02) to preserve separate stochastic source usage.

2 Conceptual Model Diagram

Vehicle lifecycle (textual flow):

South arrival at A → classify type → join A N/S queue (priority by type) → if emergency and non-N/S, request preemption → service at A → either turn east (exit) or enter A–B link (subject to blocking rules for non-emergency) → arrive at B N/S queue → service at B → turn east or exit north (throughput count).

In parallel, W→E streams at A and B queue and serve during W/E green only.

3 Experimental Design

- Run length per replication: 7,200 s (2 hours)
- Warm-up: first 900 s discarded
- Replications per scenario: 20
- Confidence interval: 95% normal approximation $\bar{x} \pm 1.96 \cdot s/\sqrt{n}$

Random streams (separate generators from one seed sequence):

- master seed: 20260407
- south arrivals, W/E arrivals at A, W/E arrivals at B
- vehicle classification
- turning at A, turning at B
- service times at A, service times at B
- link travel-time noise

4 Results

Table 1 reports key performance indicators as mean [95% CI].

Table 1: Main KPI Summary (mean [95% CI])

KPI	S1 Uncoord.	S2 Coord.	S3 ord.+Emerg.	Co-
Avg delay (through S→N) [s]	121.81 [102.58, 141.05]	96.27 [84.40, 108.15]	70.21 [61.79, 78.63]	
Avg delay (W→E) [s]	21.74 [21.38, 22.09]	21.46 [21.09, 21.84]	22.57 [22.22, 22.93]	
Throughput northbound at B [veh/h]	493.91 [487.93, 499.90]	495.94 [487.66, 504.22]	501.77 [493.45, 510.09]	
Avg link occupancy [veh]	4.30 [4.24, 4.36]	4.25 [4.20, 4.31]	4.26 [4.19, 4.32]	
Blocking frequency [fraction cycles]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	
Preemptions per replication [count]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	53.15 [50.21, 56.09]	
Recovery time after preemption [s]	NA	NA	15.79 [15.06, 16.52]	

Table 2: Queue KPI Summary (mean [95% CI])

KPI	S1 Uncoord.	S2 Coord.	S3 ord.+Emerg.	Co-
Avg queue A N/S [veh]	24.34 [19.48, 29.20]	21.29 [18.24, 24.33]	13.25 [11.19, 15.31]	
Max queue A N/S [veh]	62.10 [53.22, 70.98]	56.60 [50.05, 63.15]	41.90 [36.35, 47.45]	
Avg queue B N/S [veh]	4.61 [4.43, 4.80]	2.14 [2.01, 2.27]	3.17 [2.92, 3.42]	
Max queue B N/S [veh]	18.90 [17.82, 19.98]	13.75 [12.47, 15.03]	18.90 [17.82, 19.98]	
Avg queue A W/E [veh]	2.40 [2.32, 2.49]	2.40 [2.31, 2.49]	2.58 [2.49, 2.68]	
Max queue A W/E [veh]	12.60 [11.78, 13.42]	13.30 [12.23, 14.37]	14.10 [13.28, 14.92]	
Avg queue B W/E [veh]	2.45 [2.36, 2.53]	2.39 [2.33, 2.45]	2.48 [2.39, 2.57]	
Max queue B W/E [veh]	12.80 [12.12, 13.48]	12.90 [11.86, 13.94]	12.75 [12.01, 13.49]	

5 Comparative Discussion

Effect of coordination (S2 vs S1). Coordination reduced through-delay from 121.81 s to 96.27 s (about 20.97% reduction) and increased northbound throughput from 493.91 to 495.94 veh/h (about 0.41% increase). Link occupancy slightly decreased (about 1.09%), and W/E delay changed only marginally.

Effect of emergency preemption (S3 vs S2). In this configuration, S3 produced lower through-delay (70.21 s, additional 27.07% reduction relative to S2) and slightly higher throughput (+1.18%). W/E delay increased by about 5.16%, which is consistent with occasional interruptions of W/E service due to preemption. Mean preemptions were 53.15 per replication and mean recovery time was 15.79 s.

Most affected KPI. Based on relative change between S2 and S3, the most affected major KPI is through-delay (largest absolute percentage change among delay/throughput/link-occupancy indicators).

Interpretation note. The observed improvement in through-delay under S3 can occur because

preemptions force additional N/S service windows, effectively benefiting the corridor stream under this demand setting. This is a model-consistent outcome, not necessarily universal for all demand/preemption levels.

6 Verification and Validation

We performed sanity checks directly from replication outputs:

- Non-negativity and finite checks for queue KPIs and throughput passed.
- Blocking frequency remained within $[0,1]$ and was exactly 0.0 in all scenarios.
- Preemption count was zero in scenarios without emergencies and positive in scenario 3.
- Link occupancy statistics stayed within capacity-consistent bounds.

Additionally, we validated qualitative behavior by confirming: (i) emergency events trigger immediate preemption when N/S is not green, (ii) bus priority is non-preemptive, and (iii) the controller resumes regular cycle after emergency handling.

7 Reproducibility and Artifacts

All code and outputs are in the project notebook and CSV artifacts:

- `traffic_corridor_simulation.ipynb`
- `outputs/replication_results.csv`
- `outputs/kpi_summary_ci95.csv`
- `outputs/scenario_comparison.csv`
- `outputs/report_ready_kpi_table.csv`

Note to submitters: Replace team member names and optionally add one figure (from notebook plots) to strengthen visual comparison.